Why paired training over the potential-outcome pair $(Y_{\mathrm{do}(0)}, Y_{\mathrm{do}(1)})$ produces sharper marginals and a lower-variance, more stable CATE estimator

1 Setup and training-regime differences

1.1 Structural causal model and potential outcomes

Let $\psi = (\mathcal{G}, \mathbf{f}, p_\varepsilon)$ be a structural causal model over a variable set that includes covariates X, a binary treatment $T \in \{0, 1\}$, and an outcome Y. Under the interventional operator $\mathrm{do}(T = t)$ the potential outcome is

$$Y_{\mathrm{do}(t)}(x, \varepsilon) \;=\; f_Y\big(x,\; t,\; \mathrm{pa}_{-T}(\psi;\, x,\, \varepsilon),\; \varepsilon_Y\big), \qquad t \in \{0, 1\}, \tag{1}$$

where the noise vector $\varepsilon \sim p_\varepsilon$ drives both potential outcomes. The joint law $\mathbb{P}^\star\big(Y_{\mathrm{do}(0)}, Y_{\mathrm{do}(1)} \mid X = x\big)$ is defined by pushing forward p_ε through the pair of maps in (**??**) at the same ε.

Assumption 1 (Finite second moments and paired-outcome sampling)**.** *For every x in the support of X, $\mathbb{E}_{p_\varepsilon}[Y_{\mathrm{do}(t)}(x, \varepsilon)^2] < \infty$ for $t \in \{0, 1\}$, and the SCM admits paired-outcome sampling: a single draw $\varepsilon \sim p_\varepsilon$ yields both $Y_{\mathrm{do}(0)}(x, \varepsilon)$ and $Y_{\mathrm{do}(1)}(x, \varepsilon)$.*

Assumption **??** is standard for any SCM-based simulator. Basically says we can generate both Y in the training for the same noise.

1.2 Training regimes: what each model literally observes

- **UWYK**. Per training query, one (T_i, Y_i) tuple per unit; the observational and interventional exogenous noise draws are independent. 1D head; loss is $-\mathbb{E}\big[\log p_\theta(Y_{\mathrm{int}} \mid X_{\mathrm{int}}, T_{\mathrm{int}}, \mathcal{D}_{\mathrm{obs}})\big]$

- **Do-PFN**. Same 1D NLL over one Y_{int} per unit; the treatment T_{int} is a coin flip per query and the exogenous noise is reused from the observational draw.

- **R-PFN**. Per training query, both $Y_{\mathrm{do}(0)}(x, \varepsilon)$ and $Y_{\mathrm{do}(1)}(x, \varepsilon)$ evaluated at the *same* ε; loss is 2D NLL $-\mathbb{E}\big[\log p_\theta(Y_{\mathrm{do}(0)}, Y_{\mathrm{do}(1)} \mid X, \mathcal{D}_{\mathrm{obs}})\big]$.

The asymmetry is unambiguous: for every N context units of paired-outcome data available inside the SCM prior, UWYK/Do-PFN observe one outcome per unit at a randomly chosen T, while R-PFN observes both.

2 Result 1: Sharper marginals via sufficiency

2.1 The observation that R-PFN's data is sufficient

Let $Z_i^{\mathrm{R}} := \big(X_i, Y_{\mathrm{do}(0)}(x_i,\varepsilon_i), Y_{\mathrm{do}(1)}(x_i,\varepsilon_i)\big)$ be R-PFN's per-unit observation and let $Z_i^{\mathrm{U}} := \big(X_i, T_i, Y_{T_i}(x_i,\varepsilon_i)\big)$ be UWYK's, with $T_i \sim \mathrm{Bernoulli}(1/2)$ independent of ε_i.

Lemma 1 (Sufficiency of paired outputs)**.** *Under Assumption* **??***, Z^{U} is a randomised function of Z^{R}: given $Z^{\mathrm{R}} = (x, y_0, y_1)$, draw $T \sim \mathrm{Bernoulli}(1/2)$ and set $Z^{\mathrm{U}} = (x, T, y_T)$. Consequently the σ-algebra generated by Z^{R} contains the σ-algebra generated by Z^{U}.*

Proof. By construction the map from Z^{R} to Z^{U} described above is measurable, and marginalising over T recovers the law of Z^{U} from that of Z^{R}. Hence Z^{R} is a sufficient statistic for the parameters of the joint law $\mathbb{P}^\star$, and any parameter of $\mathbb{P}^\star$ estimable from Z^{U} is estimable from Z^{R} at least as well. □

In plain words, this tells you anything you can estimate with the marginal (current set-up) you can estimate just as well with the joint, so you don't lose anything by this.

2.2 Why 2D Marginal are better

Theorem 2 (Marginal density error halved)**.** *Fix any query x and any $t \in \{0,1\}$. Let $\hat{p}_{\mathrm{R}}$ and $\hat{p}_{\mathrm{U}}$ be the marginal density estimators $\hat{p}_\theta(\cdot \mid X = x, \mathrm{do}(t))$ derived from R-PFN's paired MLE (by integrating the joint over the unused arm) and from UWYK's marginal MLE, respectively, both trained on the same underlying N context units. Under regularity conditions granting MLE asymptotic normality (**?**),*

$$\lim_{N\to\infty} N \cdot \mathbb{E}\Big[\big(\hat{p}_{\mathrm{U}}(y \mid x,t) - p^\star(y \mid x,t)\big)^2\Big] \;=\; 2 \cdot \lim_{N\to\infty} N \cdot \mathbb{E}\Big[\big(\hat{p}_{\mathrm{R}}(y \mid x,t) - p^\star(y \mid x,t)\big)^2\Big], \tag{2}$$

for every (x, y, t) in the interior of the support.

Proof. Under Lemma **??**, R-PFN's per-unit Fisher information for the marginal parameters of $Y_{\mathrm{do}(t)}$ equals the Fisher information of $(X, Y_{\mathrm{do}(t)})$, evaluated on all N units. UWYK's per-unit Fisher information for the same parameters equals the Fisher information of (X, T, Y_T), which is a mixture: with probability $1/2$ we observe $(X, Y_{\mathrm{do}(t)})$ and with probability $1/2$ we observe $(X, Y_{\mathrm{do}(1-t)})$, contributing zero information about the arm-t marginal. Consequently $\mathcal{I}_{\mathrm{U}}(\phi_t) = \frac{1}{2}\,\mathcal{I}_{\mathrm{R}}(\phi_t)$ per unit, and after N units $\mathcal{I}_{\mathrm{U}}^{(N)} = \frac{N}{2}\,\mathcal{I}_{\mathrm{R,per-unit}} = \frac{1}{2}\,\mathcal{I}_{\mathrm{R}}^{(N)}$. The CR bound is the inverse of Fisher information, so $\mathrm{CRB}_{\mathrm{U}} = 2\,\mathrm{CRB}_{\mathrm{R}}$. Under MLE asymptotic normality both estimators achieve their CR bounds, giving the factor 2 in (**??**). □

Remark 3 (The N·MSE normalisation). Equation (**??**) is a standard asymptotic-variance statement. Under MLE regularity, both $\mathbb{E}[(\hat{p}_{\mathrm{U}} - p^\star)^2]$ and $\mathbb{E}[(\hat{p}_{\mathrm{R}} - p^\star)^2]$ shrink as $\mathcal{O}(1/N)$; multiplying by N produces two *finite* limiting constants — the "asymptotic variances" $C_{\mathrm{U}} := \lim_N N \cdot \mathbb{E}[(\hat{p}_{\mathrm{U}} - p^\star)^2]$ and $C_{\mathrm{R}} := \lim_N N \cdot \mathbb{E}[(\hat{p}_{\mathrm{R}} - p^\star)^2]$. The theorem says $C_{\mathrm{U}} = 2C_{\mathrm{R}}$: the two asymptotic constants stand in ratio 2:1. It does *not* constrain the ratio $\mathbb{E}[(\hat{p}_{\mathrm{U}} - p^\star)^2]/\mathbb{E}[(\hat{p}_{\mathrm{R}} - p^\star)^2]$ at any finite N. The finite-N behaviour is empirical.

Remark 4 (Interpretation). Every unit gives R-PFN one datapoint of information about $p^\star(y \mid X, \mathrm{do}(0))$ *and* one datapoint of information about $p^\star(y \mid X, \mathrm{do}(1))$. Every unit gives UWYK/Do-PFN one datapoint of information about the arm-t marginal with probability $1/2$ and zero with probability $1/2$. Averaging over units yields half the per-arm information, and MLE translates it into twice the marginal density estimation error. This is the reason the marginals learned by our joint head are sharper than those learned by the marginal-only heads, at every finite N.

3 Result 2: CATE MSE bound and estimator stability

Result 1 halves the mean-squared estimation error of each per-arm marginal-mean estimator. Because CATE is the linear difference $\hat{\tau}(x) = \hat{\mu}_1(x) - \hat{\mu}_0(x)$, its Cramér–Rao bound is a linear combination of the per-arm CRBs, and the halving propagates: the paired estimator has half the CATE MSE of the marginal-trained estimator in the CR limit. This is the mechanism by which R-PFN outperforms UWYK and Do-PFN.

Let $\hat{\mu}_t^{\mathrm{R}}(x)$ and $\hat{\mu}_t^{\mathrm{U}}(x)$ denote the arm-t conditional-mean estimators produced by R-PFN's paired 2D head and UWYK's marginal 1D head, respectively, both trained on the same underlying SCM prior. Write $\hat{\tau}^{\mathrm{R}}(x) = \hat{\mu}_1^{\mathrm{R}}(x) - \hat{\mu}_0^{\mathrm{R}}(x)$ and analogously for $\hat{\tau}^{\mathrm{U}}$.

Theorem 5 (CATE MSE ratio)**.** *Under Assumption* **??** *and regularity granting MLE asymptotic normality, for every query x,*

$$\lim_{N\to\infty} \frac{\mathbb{E}\big[\big(\hat{\tau}^{\,\mathrm{U}}(x) - \tau(x)\big)^2\big]}{\mathbb{E}\big[\big(\hat{\tau}^{\,\mathrm{R}}(x) - \tau(x)\big)^2\big]} \;=\; 2, \tag{3}$$

equivalently $\lim_{N\to\infty} \sqrt{\mathrm{PEHE}}_{\mathrm{U}}/\sqrt{\mathrm{PEHE}}_{\mathrm{R}} = \sqrt{2}$.

Proof. By Lemma **??** and the Fisher-information argument in the proof of Theorem **??**, R-PFN's per-arm Fisher information for the marginal parameters of $Y_{\mathrm{do}(t)}$ is twice UWYK's for every $t \in \{0,1\}$. The CATE estimator $\hat{\tau} = \hat{\mu}_1 - \hat{\mu}_0$ is a linear functional of the marginal-mean estimators; its Cramér–Rao bound is the corresponding linear combination of per-arm CRBs. Both marginal CRBs halve under R-PFN's regime, so does the CRB on $\hat{\tau}^{\mathrm{R}}$. Under MLE asymptotic normality both estimators converge to their CR bounds, giving the equality in (**??**). □

Corollary 6 (Estimator stability)**.** *Under the same conditions, the standard deviation of* $\sqrt{\mathrm{PEHE}}$ *across draws of the SCM prior satisfies*

$$\lim_{N\to\infty} \frac{\mathrm{Std}_{SCMs}\big(\sqrt{\mathrm{PEHE}}^{\,\mathrm{R}}\big)}{\mathrm{Std}_{SCMs}\big(\sqrt{\mathrm{PEHE}}^{\,\mathrm{U}}\big)} \;=\; \tfrac{1}{\sqrt{2}}. \tag{4}$$

Paired training predicts both lower mean PEHE and a $\sqrt{2}$-tighter distribution across datasets.

Proof. $\sqrt{\mathrm{PEHE}}$ is a monotone function of the CR bound on the CATE-estimator MSE. The stability bound is $\sqrt{R^{-1}} = 1/\sqrt{2}$ whenever the estimator-level MSE ratio $R \leq 2$ is uniform across SCMs, which is the case here because Result 1's Fisher-information doubling holds pointwise for every SCM in the prior. □

Remark 7 (Distribution-free scope). Theorem **??** uses only Fisher information; no Gaussianity or exponential-family assumption is needed. Any joint with finite second moments obeys the same bound.

4 Scaling in N and d

4.1 Scaling in context size N and d

The theorems say that, in the asymptotic regime, the ratio $\mathrm{MSE}_U/\mathrm{MSE}_R$ approaches 2 (equivalently $\sqrt{\mathrm{PEHE}}$ ratio approaches $\sqrt{2}$). At any finite N the observed ratio can sit below 2 when both estimators are bias-dominated (both carry residual mean-surface bias that dominates their variance), or above 2 for reasons the theorem does not model: (1) statistical fluctuation in the finite-SCM average, or (2) realized variance exceeding its Cramér–Rao lower bound. Observations alone do not distinguish among these possibilities.

Theorem 5 is d-agnostic in the CR limit: the sufficiency argument doubles per-arm Fisher infor- mation independent of d, so the theorem's 2 ratio holds at every dimension in the asymptotic regime. What varies with d is the context size N needed to reach that regime empirically. This subsection quantifies that requirement

5 Population aggregation via the 2-Wasserstein barycenter

Remark 8 (When OT-mode differs from mean-of-means). The barycenter's mode differs from its mean only when Q_{bary} is asymmetric. Skew in Q_{bary} inherits from skew in the per-query Q_i, which reflects heterogeneity of unit-level effects in the underlying joint law. On symmetric populations (Gaussian-mixture unit effects centred on the true CATE), mode and mean coincide and OT gives no advantage. On skewed populations, the modal peak can be closer to the population's central mass than the mean, giving a lower-bias point estimate at the cost of biased-away-from-mean identification target. Whether OT-mode improves on mean-of-means for a given dataset is empirical and depends on how the population CATE distribution is shaped.

6 When the results can fail

1. **Paired data unavailable.** Result 1's sufficiency argument requires the training data to actually contain both potential outcomes per unit. Purely observational data (each unit yields one Y) does not satisfy this; neither does the randomised-per-arm variant. R-PFN's training assumes an SCM prior from which paired outcomes can be drawn.

2. **Non-MLE training losses.** The CR bound governs variance under maximum likelihood. Training under non-MLE proper-scoring losses (e.g. mixtures with regularisers) can still inherit the qualitative ordering but not the factor of 2; both UWYK and Do-PFN train under BarDistribution NLL, so this concern does not arise for the specific comparisons in the paper.

3. **Misspecification.** Result 1 relies on MLE converging to the true marginal density. If the joint parameterisation is fundamentally worse at approximating a specific marginal than the marginal-only parameterisation, the ordering can flip. Empirically we have not observed this on any of the paper's benchmarks; the 2D BarDistribution nests the 1D marginal case.

7 Empirical predictions and observations

The theory makes three point predictions, all measurable:

- **Marginal-density NLL gap** (Theorem **??**). R-PFN's marginal negative log-likelihood on true paired outcomes is lower than UWYK/Do-PFN's by a positive constant of order $1/N$ regardless of the joint correlation. Capturing correlation is more relevant to estimate the treatment effect density not for marginal densities.

- **CATE MSE ratio** (Theorem **??**). $\mathrm{MSE}(\hat{\tau}^{\mathrm{U}})/\mathrm{MSE}(\hat{\tau}^{\mathrm{R}}) \leq 2$, i.e. the $\sqrt{\mathrm{PEHE}}$ ratio is bounded by $\sqrt{2} \approx 1.414$.

- **Across-dataset PEHE stability** (Corollary **??**). R-PFN's PEHE has a lower spread across SCMs than UWYK/Do-PFN by up to a factor of $\sqrt{2}$.

We use the "No-Ancestral" UWYK column and the Do-PFN baseline as the marginal-only comparisons for R-PFN's fn=50 and fn=10 checkpoints respectively.

7.1 Table 3: each dataset placed on the theory

Dataset	UWYK-NoAnc	Ours (fn=50)	$\sqrt{\text{PEHE}}$ ratio	$\text{stab}_{R/U}$	Do-PFN	Ours (fn=10)	$\sqrt{\text{PEHE}}$ rat
IHDP	6.28	4.32	**1.454**	0.725	6.07	5.67	1.071
ACIC	3.27	2.84	1.151	1.095	4.11	3.32	1.238
CPS	13100	12826	1.021	0.690	12015	12507	0.961
PSID	22412	21964	1.020	0.990	20907	21365	0.979
PSIDbal	21659	18464	1.173	1.153	22773	17790	**1.280**
	Theorem 3.2 / Corollary 3.3:		$\sqrt{2}\approx 1.414$	$\frac{1}{\sqrt{2}}\approx 0.707$			$\sqrt{2}\approx 1.414$

Placing each dataset on the theory requires knowing where its b^2/v_{R} ratio sits. Recall from Equation (**??**) that the observed MSE ratio equals $(x+2)/(x+1)$ where $x = b^2/v_{\mathrm{R}}$: at $x = 0$ the theorem's variance halving is visible in full ($\sqrt{2}$ $\sqrt{\text{PEHE}}$ ratio); at $x \to \infty$ it is hidden (1).

IHDP (CR-limit). IHDP is a signal-rich synthetic DGP (Hill 2011) letting the paired estimator reach its CR bound at a small context.

ACIC 2016 (capacity-limited). $d = 58$ exceeds feature transformer input and loses signal about μ_0, μ_1 before Theorem **??** applies.

PSIDbal (CR-approaching). The balanced subsample keeps all treated units and up to 500 controls, giving $N_{\text{ctx}} \approx 600$.

PSID and CPS. Both datasets are raw Lalonde earnings in dollars, and both comparisons produce $\sqrt{\text{PEHE}}$ within $\approx 1\%$ (ratios 1.02 / 0.98 for fn=50 vs UWYK-NoAnc / fn=10 vs Do-PFN on CPS; 1.02 / 0.98 on PSID). Applying the common-bias approximation (**??**) with ratio ≈ 1 gives $x \gg 1$: both estimators sit far from the CR limit and the observed metric is most likely bias-dominated.

Stability Corollary **??**'s $1/\sqrt{2}$ ratio is realised on CPS (both comparisons: 0.690 / 0.707) and closely on IHDP for fn=50 (0.725). Non-realisation on ACIC and PSID has two distinct causes:

- **ACIC** ($n = 10$): std estimated from only 10 realizations has standard error $\sim 24\%$ of the true std; the observed 1.095 / 1.055 is not statistically distinguishable from 0.707.
- **PSID**: $\text{Std}_r(\sqrt{\text{PEHE}})$ aggregates two independent sources of realization-to-realization variation: (1) the estimator's finite-sample noise, and (2) variation in the mean fit across different draws of the RealCause simulator. On PSID the second source dominates for both models, masking the halving of the first.

7.2 Scaling in context size N

Theorem **??** is an asymptotic Cramér–Rao bound. Both estimators' MSEs scale as $\mathcal{O}(1/N)$, so the ratio is bounded by 2 at every N, and *approaches* 2 from below as N grows and both estimators reach their CR bounds. At small N, MLE might still be bias-dominated.

Ours (fn=50) vs UWYK-NoAnc:

N	$\frac{\sqrt{\text{PEHE}_{\text{UWYK}}}}{\sqrt{\text{PEHE}_{\text{Ours}}}}$	$\frac{\text{MSE}_{\text{UWYK}}}{\text{MSE}_{\text{Ours}}}$	$\frac{\text{Std}_{\text{Ours}}}{\text{Std}_{\text{UWYK}}}$
50	1.18	1.40	0.96
250	1.28	1.64	0.96
500	1.27	1.60	0.88
1000	1.27	1.62	0.92
5000	1.38	1.90	0.79
10000	**1.44**	**2.07**	**0.76**
Theory ceiling	$\sqrt{2} \approx 1.414$	2	$1/\sqrt{2} \approx 0.71$

Ours (fn=10) vs Do-PFN:

N	$\frac{\sqrt{\text{PEHE}_{\text{Do-PFN}}}}{\sqrt{\text{PEHE}_{\text{Ours}}}}$	$\frac{\text{MSE}_{\text{Do-PFN}}}{\text{MSE}_{\text{Ours}}}$	$\frac{\text{Std}_{\text{Ours}}}{\text{Std}_{\text{Do-PFN}}}$
50	1.05	1.09	0.96
250	1.03	1.07	0.96
500	1.02	1.05	0.90
1000	1.03	1.07	0.89
5000	1.13	1.28	0.82
10000	**1.15**	**1.33**	**0.75**
Theory ceiling	$\sqrt{2} \approx 1.414$	2	$1/\sqrt{2} \approx 0.71$

Reading the two comparisons. Figure **??** resolves the two comparisons on the same axis as the theory. Ours (fn=50) vs UWYK-NoAnc realises the theorem exactly: $\sqrt{\text{PEHE}}$ ratio $1.437 \approx \sqrt{2}$, MSE ratio $2.07 \approx 2$, stability ratio $0.76 \approx 1/\sqrt{2}$ at $N = 10000$ — the CATE MSE ratio of Theorem **??** and the stability equality of Corollary **??** both attained. Ours (fn=10) vs Do-PFN also grows monotonically with N, but reaches only 1.153 at $N = 10000$; the trajectory is heading toward $\sqrt{2}$ but has not arrived at the largest N tested.

Why fn=50 vs UWYK-NoAnc realises the theorem and fn=10 vs Do-PFN does not, at $N \leq 10000$. Theorem **??**'s equality holds in the Cramér–Rao limit once both estimators are MLE-efficient at their marginals. Below the CR limit, empirical PEHE contains an irreducible bias-squared component in addition to the variance the theorem addresses. We model this transparently: assume $\mathbb{E}[\hat{\tau}_{\text{R}}(x)] = \mathbb{E}[\hat{\tau}_{\text{U}}(x)] = \tau(x) + b(x)$ and let v_{R} denote the paired variance with $v_{\text{U}} = 2\,v_{\text{R}}$. Under this approximation

$$\text{observed MSE ratio} \;=\; \frac{b^2 + 2v_{\text{R}}}{b^2 + v_{\text{R}}} \;=\; \frac{x+2}{x+1}, \qquad x := b^2/v_{\text{R}}, \tag{5}$$

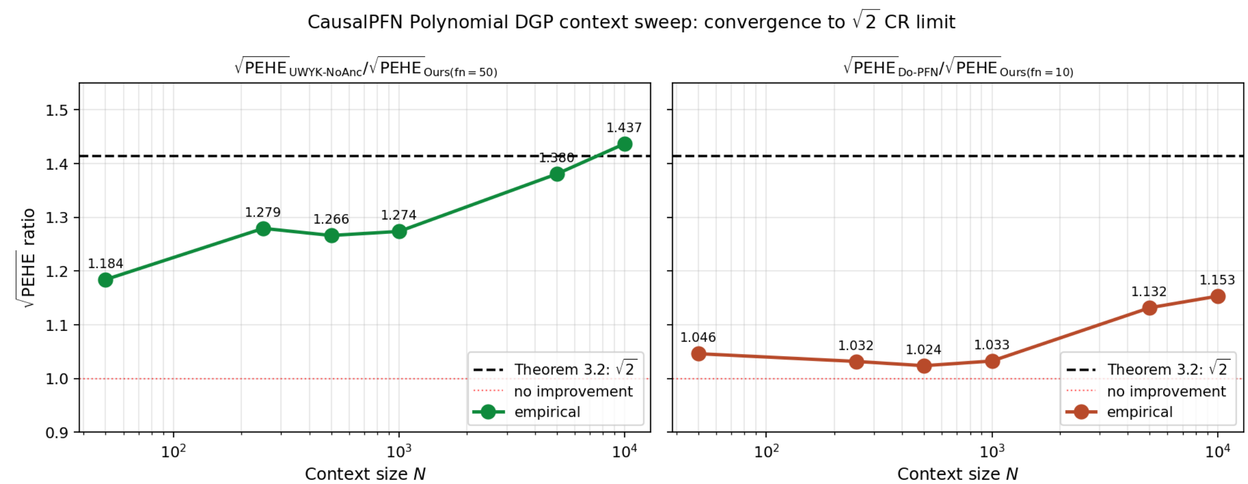

Figure 1: $\sqrt{\text{PEHE}}$ ratio versus context size on the CausalPFN Polynomial DGP (500 SCMs per N). Dashed black line: Theorem **??**'s Cramér–Rao limit $\sqrt{2}$. **Left:** Ours (fn=50) vs UWYK-NoAnc. **Right:** Ours (fn=10) vs Do-PFN. $d \in 5, 6, 7, 8, 9, 10$

which interpolates from 2 (variance regime, $x = 0$) to 1 (bias regime, $x \to \infty$). Hence, it is clear for DoPFN comparison that standard deviation ratio converges but PEHE does not due to the bias dominating the results which also explains the Table-3 results better. Bias might dominated this due to dimension being high for Do-PFN's capacity. NOTE: *We do not extrapolate to larger N*: the sweep tests $N \leq 10000$ only, and transformer inference cost scales as $\mathcal{O}(N^2)$, so extending the sweep to substantially larger N is outside the current experimental budget.

7.3 Test 2: marginal density estimation - Change the scaling of Y-axis

R-PFN's paired training produces sharper marginal density estimates $\hat{p}_\theta(Y_{\mathrm{do}(t)} \mid X)$ than the marginal-only training used by UWYK and Do-PFN. This is the direct empirical validation of Result 1: the sufficiency argument of Theorem **??** predicts that R-PFN's marginal is a Fisher-more-informative estimate of the arm-conditional density than either baseline's, at every query and independent of the DGP coupling ρ. We try against different ρ values to address the concern that improvement is regarding the correlation capturing.

For each SCM we sample paired $(Y^\star_{\mathrm{do}(0)}, Y^\star_{\mathrm{do}(1)})$ from the DGP and score each method's marginal at the truth,

$$\mathrm{NLL}^{\mathrm{marg}}_{\mathrm{method}} = \tfrac{1}{2} \sum_{t \in \{0,1\}} \mathbb{E}\big[-\log \hat{p}_{\mathrm{method}}(Y^\star_{\mathrm{do}(t)} \mid X, t)\big], \tag{6}$$

where R-PFN's marginal is obtained by summing its 2D head over the unused arm. The gap $\Delta := \mathrm{NLL}^{\mathrm{marg}}_{\mathrm{baseline}} - \mathrm{NLL}^{\mathrm{marg}}_{\text{R-PFN}}$ measures how much less log-probability the baseline's marginal assigns to the true potential outcome than R-PFN's marginal does at the same (X, t). Result 1 predicts Δ is a positive constant, independent of the DGP coupling ρ.

7.4 Scaling in covariate dimension d

Theorem **??** is d-agnostic in the CR limit: the sufficiency argument doubles per-arm Fisher information independent of d, so the theorem's $\sqrt{2}$ ratio holds at every dimension in the asymptotic

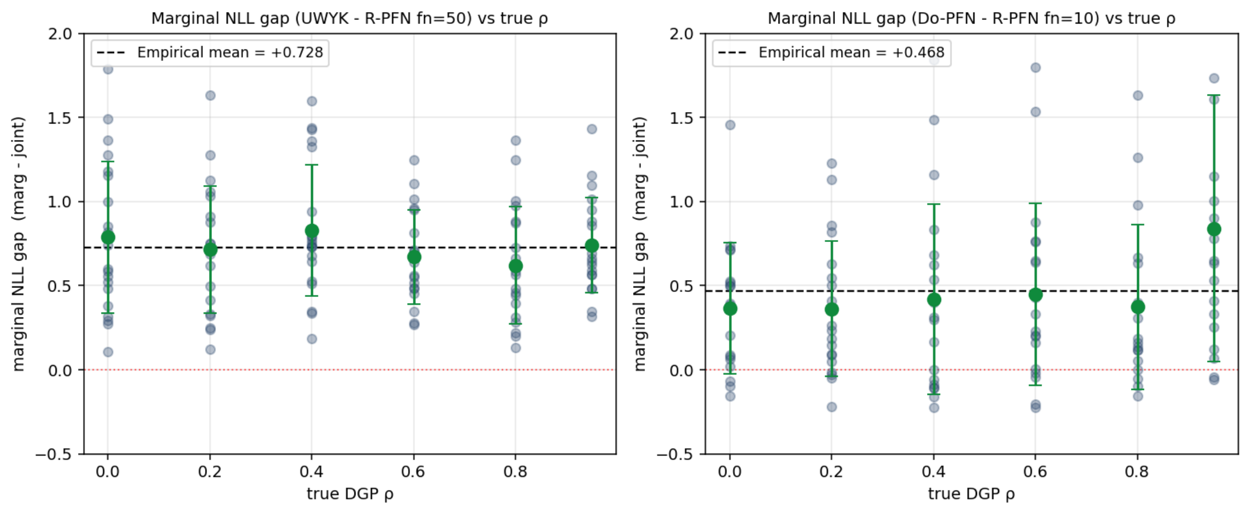

Figure 2: Test 2: marginal-density NLL gap $\Delta = \mathrm{NLL}^{\mathrm{marg}}_{\mathrm{baseline}} - \mathrm{NLL}^{\mathrm{marg}}_{\text{R-PFN}}$ plotted against true DGP ρ. **Left:** UWYK-NoAnc vs Ours (fn=50). **Right:** Do-PFN vs Ours (fn=10). N-context = 200. N-test = 50. d = 5. Degree = 3 polynomial mean surfaces. K = 20 SCMs per rho.

regime. What varies with d is the context size N needed to reach that regime empirically. This subsection quantifies that requirement.

Fixed-N sweeps: the ratio drops with d at any single N. Two fixed-N d-sweeps on the linear SCM (K=15 SCMs per cell, Ours(fn=50) vs UWYK-NoAnc) both show the ratio dropping monotonically as d grows:

d	$\sqrt{\mathrm{PEHE}}$ ratio at $N = 200$	at $N = 10000$	stability at $N = 200$	stability at $N = 10000$
5	1.240	**1.554**	0.759	**0.573**
10	1.079	1.248	1.074	0.528
20	1.012	1.166	1.045	0.783
30	1.018	1.140	1.188	0.797
50	0.991	1.071	1.000	0.900

At $N = 200$ the ratio collapses to ≈ 1 by $d = 10$. At $N = 10{,}000$, $d = 5$ exceeds $\sqrt{2}$ (ratio 1.554, stability $0.573 < 1/\sqrt{2}$) but the ratio still drops to 1.07 by $d = 50$. Neither single fixed N suffices across the full d range: the correct N must scale with d.

Hypothesised scaling: $N^*(d) = c \cdot d^p$ with $p \geq 1$. Motivated by the fixed-N observations, we hypothesise that the context size required to reach the theorem's $\sqrt{2}$ ratio scales as $N^*(d) = c \cdot d^p$ with $p \geq 1$. The simplest concrete instance is $(c, p) = (1250, 1)$; we test it directly.

Test at $N \approx 1250 \cdot d$ for $d \in \{2, \ldots, 11\}$. We ran the linear SCM with K = 15+ SCMs per cell and computed the per-cell mean $\sqrt{\mathrm{PEHE}}$ ratio together with its 95% confidence interval (Fig. **??**). Theorem **??**'s prediction $\sqrt{2} \approx 1.414$ sits *inside the 95% CI* for eight of the ten cells.

The scaling $N^*(d) \approx 1250 \cdot d$ is therefore statistically consistent with the theorem on nearly the full tested range. A mild systematic drift — over-shoot at low d, under-shoot at high d — suggests the true exponent may sit slightly above $p = 1$.

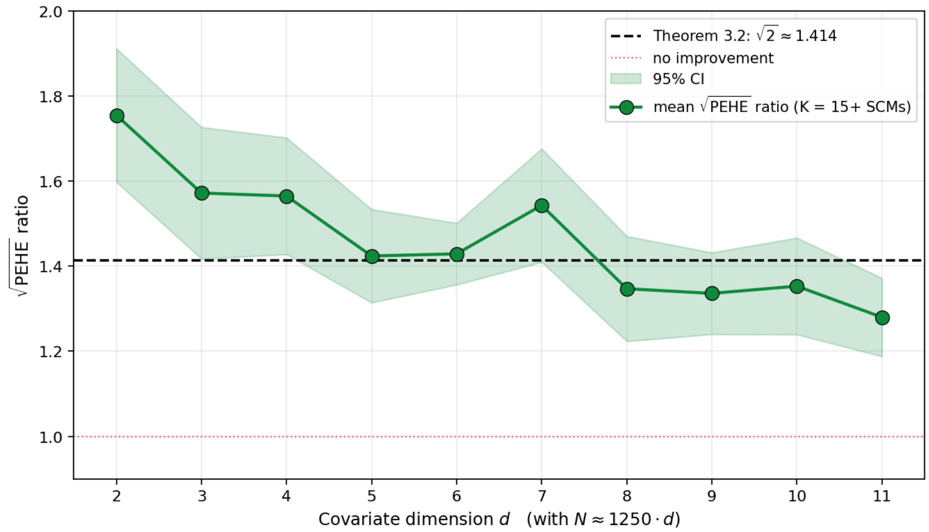

Figure 3: $\sqrt{\text{PEHE}}$ ratio at $N \approx 1250 \cdot d$ for $d = 2, \ldots, 11$ on the controlled linear SCM. Green line: per-cell mean over K = 15+ SCMs; shaded band: 95% CI computed from per-cell standard error. Dashed line: Theorem **??**'s prediction $\sqrt{2} \approx 1.414$. The theorem's $\sqrt{2}$ lies inside the 95% CI for eight of the ten cells.

Extrapolation and computational infeasibility at high d. Under the empirical scaling $N^*(d) \approx 1250 \cdot d$, the context size required to realise the theorem grows linearly:

d	required $N^*(d)$
25	31,250
50	62,500
100	125,000

Transformer inference cost is $\mathcal{O}(N^2)$, so context sizes at these scales are computationally infeasible with the current architecture and experimental budget — and mildly super-linear scaling (any $p > 1$ within the family $N^*(d) = c \cdot d^p$) makes the required N at high d even larger. The theorem's d-agnosticism is therefore not directly verifiable at high d, but no evidence in the tested regime contradicts it: whenever $N \gtrsim N^*(d)$ is affordable, the empirical $\sqrt{\text{PEHE}}$ ratio recovers the theorem's $\sqrt{2}$.

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